App Note – HVLP + D8 Motor Temperature Sensor Configuration

DOCUMENT INFORMATION

Document Reference	
Document Status	Draft
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Revision	2.0
Issue Date	23rd August 2021
Archive Location	

REVISION HISTORY

Revision	Date	Details	Author
1.0	2015-04-23	Initial version	PS
2.0	2021-08-23	Update template	DH

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1 Introduction

1.1 Purpose and Scope

HVLP and Dragon8 configuration of motor thermistors.

1.2 Related documents

Ref	Doc Ref	Title	Origin	Revision
1				
2				

Table 1. Related Documents

1.3 Document Conventions and Terminology

Typeface	Indicates
WARNING	Very important information for the document user. Use this to indicate instructions or constraints the document user must comply with to prevent serious problems.
CAUTION	Important information for the document user. Use this to indicate instructions or constraints the document user must comply with to prevent undesirable problems.
NOTE	Useful information for the document user.

Table 2. Typographical Conventions

1.4 Glossary

Term	Description

Table 3. Glossary

1.5 Conditions for document usage

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2 Overview

The following block diagram shows the basic overview of motor temperature thermal cutback:

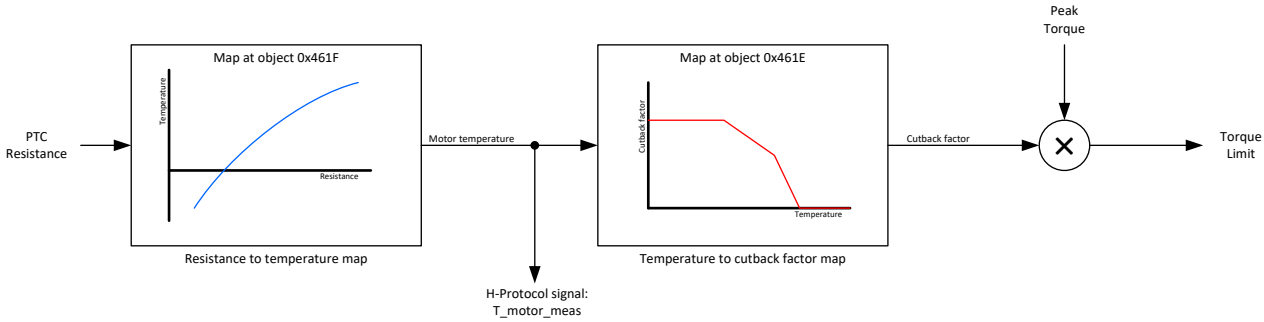


Figure 1 - Overview of operation of motor thermal cutback

Broadly speaking, the following actions occur in sequence to calculate an output torque limit:

- PTC resistance is calculated from the PTC analogue input, based on the supply voltage to the PTC, the voltage measured from the PTC, and characteristics of the internal measurement circuit.
- The PTC resistance is converted to a temperature, using a programmable map of resistances to temperatures, set at object 0x461F
- The temperature is converted to a cutback factor, using a programmable map of temperature to cutback factor, set at object 0x461E
- Peak torque is multiplied by the cutback factor, resulting in the torque limit. Both drive and brake output torque will be limited to this value

Programmable maps at objects 0x461E and 0x461F allow a range of temperature sensors to be configured, with varying cutback profiles.

2.1 Observations and Limitations

The cutback scheme works by limiting motor torque. Ideally, the cutback scheme would limit current to the motor, since it is current flow that heats the motor, not output torque.

Unfortunately, limiting the output current brings about complications in the control scheme. For instance, if this is a PMAC motor that is rotating quickly and requires field weakening to prevent excessive back EMF voltages damaging other components, then there is a problem. This is due to a conflict where field weakening required current to flow (it will be asking for negative current in the d-axis), while thermal cutback is requiring less current to flow. The question of prioritising current limits then needs to be answered. There are other design problems as well, since the thermal cutback algorithms would require knowledge of the operation of the motor control – these are areas of functionality that would be better separated.

However, for most purposes, cutting back output torque is adequate, since this is tightly coupled to current flow, and cutting back one usually results in a cutback of the other. This also means that the same torque cutback scheme can be applied generically to all motor control schemes.

Nevertheless, the vehicle integrator should be aware of this limitation, and take steps such as:

- For induction motors, make sure that the fluxing current for zero torque is not so high as to result in substantial heating of the motor at zero torque demand
- For PMAC motors, ensure that field weakening currents that would normally be experienced at high speed are either not sustained for excessive periods of time, or will not result in excessive heating, during typical over-temperature conditions.

(The above list is not an exhaustive list of checks. Each application must be analysed on its own merits and limitations.)

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3 Motor PTC Configuration

The table at object 0x461F should be configured to map PTC resistance to temperature. Conversion is performed using standard linear interpolation between 8 user configurable points.

The resistance and temperature for each point must be set.

With the correct PTC profile configured, it should be possible to read motor temperature from signal `T_motor_meas` in the H-Protocol.

3.1 Example KYT84/130 and KTY84/150 configuration

The KTY84/130 and HTY84/150 temperature sensors have the following profile:

Resistance (ohms)	Temperature (deg C)
391	-30
574	18.37209
826	70
1127	120
1482	170
1893	220
2553	291.3415
2624	300

Both sensors have the same profile, however the KTY84/150 is a more accurate version of the KTY84/130.

Scaling this for the object dictionary results in the following raw values as read by the DVT:

```
dvt(101) % sdo_ro 1 0x461F
0x461F,0: 0x10 (16)
0x461F,1: 0x0187 (391)
0x461F,2: 0xffe2 (65506)
0x461F,3: 0x023e (574)
0x461F,4: 0x0012 (18)
0x461F,5: 0x033a (826)
0x461F,6: 0x0046 (70)
0x461F,7: 0x0467 (1127)
0x461F,8: 0x0078 (120)
0x461F,9: 0x05ca (1482)
0x461F,10: 0x00aa (170)
0x461F,11: 0x0765 (1893)
0x461F,12: 0x00dc (220)
0x461F,13: 0x09f9 (2553)
0x461F,14: 0x0123 (291)
0x461F,15: 0x0a40 (2624)
0x461F,16: 0x012c (300)
```

3.2 Example KTY84/151 configuration

The KTY84/151 temperature sensor has the following profile:

Resistance (ohms)	Temperature (deg C)
381	-30
566	20
806	70
1099	120
1445	170

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1831	218.2353
2495	292.0253
2558	300

Scaling this for the object dictionary results in the following raw values as read by the DVT:

```
dvt(101) % sdo_ro 1 0x461F
0x461F,0: 0x10 (16)
0x461F,1: 0x017d (381)
0x461F,2: 0xffe2 (65506)
0x461F,3: 0x0236 (566)
0x461F,4: 0x0014 (20)
0x461F,5: 0x0326 (806)
0x461F,6: 0x0046 (70)
0x461F,7: 0x044b (1099)
0x461F,8: 0x0078 (120)
0x461F,9: 0x05a5 (1445)
0x461F,10: 0x00aa (170)
0x461F,11: 0x0727 (1831)
0x461F,12: 0x00da (218)
0x461F,13: 0x09bf (2495)
0x461F,14: 0x0124 (292)
0x461F,15: 0x09fe (2558)
0x461F,16: 0x012c (300)
```

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4 Cutback Profile Configuration

Assuming we are now able to measure the motor temperature correctly, we are now able to define a cutback profile. Traditionally, on Gen4, the cutback profile was limited to cutting back over a range of 10 degrees C. However, the cutback table in object 0x461F replaces this, and allows torque to be cut back over any temperature range.

A 6 –point table is provided at object 0x461E, which allows the profile to be set. An example table, which cuts back over the range 110 to 120 deg C, would be as follows:

Temperature (deg C)	Cutback factor (p.u.)
110	1.0
112	0.8
114	0.6
116	0.4
118	0.2
120	0.0

The raw representation in the controller would be as follows – in particular, note the Q4.12 format that is used for the cutback factors:

```
dvt(101) % sdo_ro 1 0x461E
0x461E,0: 0x0c (12)
0x461E,1: 0x006e (110)
0x461E,2: 0x1000 (4096)
0x461E,3: 0x0070 (112)
0x461E,4: 0x0ccc (3276)
0x461E,5: 0x0072 (114)
0x461E,6: 0x0999 (2457)
0x461E,7: 0x0074 (116)
0x461E,8: 0x0666 (1638)
0x461E,9: 0x0076 (118)
0x461E,10: 0x0333 (819)
0x461E,11: 0x0078 (120)
0x461E,12: 0x0000 (0)
```

If cutback is desired over a different range, modifying the temperature values, which are located at odd numbered sub-indices, would be a simple change to make.

The following limitations must be observed; otherwise a dynamic range error will occur:

- Temperature values must increase progressively through the table
- Cutback factors must decrease progressively through the table
- The first cutback factor must by 1.0
- The last cutback factor must be 0.0

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5 Wire-off Protection

The system can be configured to behave differently under wire-off conditions.

Normally, the thermal cutback factor calculated for the motor is slewed internally, such that it takes 5 seconds to ramp from zero to full cutback, and vice versa. This prevents noise from the sensor appearing on the torque limit, which could result in a poor drive feel. However, in wire-off conditions, the cutback factor will ramp down to zero, but can be programmed to do so over a longer period of time. This time period can be configured at object 0x4620,6.

A typical configuration might be to have the cutback factor ramp to zero over a period of 2 minutes. In which case, this can be achieved by setting object 0x4620,6 to a value of 0x0780, which is equivalent to 120 seconds in Q12.4 format.

The sensor is defined as being in the wire-off state when the measured resistance os outside of the range defined in the table set at object 0x461F. These tables should be adjusted as necessary is an application requires a specific wire-off detection characteristics.

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